

# **INTERNSHIP**

## **Weekly Report**

**Week No: 5**

**Date: 08-01-2025**

**Class: AIML-A**

**Batch No: A-16**

**Title of the Project:** End-to-End Portfolio Management and Tax Tracking System

| <b>S.No.</b> | <b>Regd.No.</b> | <b>Name of the Student</b> | <b>Remarks / Suggestions (To be filled by Guide)</b> |
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### **Work progress :**

- Shifted focus to Signal Processing, specifically learning the difference between Frequency (Sine waves) and "Sequency" (Square waves) analysis.
- Studied the Walsh-Hadamard Transform to understand how it can be used to distinguish between a mean-reverting signal (high sequency) and a "falling knife" trend (low sequency).
- Researched Deep Learning architectures for time series, comparing LSTMs vs. Temporal Convolutional Networks (TCNs).
- Learned about "Dilated Convolutions" and "Causal Padding" to understand how TCNs can model long-term history without "cheating" (data leakage) by looking into the future.

**Signature of Guide with date**